ADDENDA FROM THE INTERACTIVE SERIES

Findings from demos 25–32, keyed to their insertion points in *Double-Slit Geometry: Definitions, Axioms, and Theorems* and *The Seam*. Everything here arose while building the diagrams; each item is verified to machine precision by the check script named at its close, in the same discipline as `geometry_checks.py`. Conventions and the calibrated rig (screen $z = 0$, ℓ_1 vertical at $Zv = 150$, ℓ_2 horizontal at $Zh = 250$, $D = Zh - Zv$) are those of the main document.

A1. Theorem 3, addition to the commentary: case 3 is the limit of case 1

The cross is not a third kind of image; it is the hyperbola collapsing onto its own asymptotes. In the screen chart the chain of a line m through E_1 with direction (Dx, Dy, Dz) has the exact coefficients

$$a = r_1 q_0 - r_0 q_1, \quad b = r_0 p_1 - r_1 p_0, \quad c = s_1 q_0 - s_0 q_1, \quad d = s_0 p_1 - s_1 p_0,$$

with $p_1 = -Dx \cdot Zv$, $p_0 = -X_1 \cdot Zv$, $q_1 = Dz$, $q_0 = Z_1 - Zv$, $r_1 = -Dy \cdot Zh$, $r_0 = -Y_1 \cdot Zh$, $s_1 = Dz$, $s_0 = Z_1 - Zh$ (obtained by composing the two Möbius maps $u \mapsto x(u)$ and $u \mapsto y(u)$; no fitting involved). Two consequences.

First, $c = Dz \cdot (Zh - Zv)$ identically, which sharpens case 2: the image is straight exactly when the line is parallel to the screen — the affine face of "meets the common transversal at screen infinity," since the direction plane of the screen is the common transversal at infinity of both slits.

Second, impose the case-3 condition that m meet ℓ_1, i.e. $X_1 Dz = Dx(Z_1 - Zv)$. Then the vertical asymptote $-d/c$ tends to $-Zv \cdot Dx/Dz$, which is exactly the collapsed pointwise trace of m (every point of m other than the touch point shares the single second coordinate $t_0 = Dx \cdot D/Dz$), and the horizontal asymptote a/c tends to $(Zh/D) \cdot s_0$, the screen trace of the S-pencil at the touch point. The two lines of the cross are the two asymptotes, arrived at continuously. The demo shows the collapse live; the identities hold to nine decimals in `check_bowing.js`, and the constancy of t along a touched line is exact.

A2. Theorem 6b, implementation note: the co-chained test is

universal, not existential

Transporting the chain criterion to 3D invites a mistake worth recording. The transversals of three pairwise non-parallel rays r_1, r_2, r_3 form one ruling of the quadric of Theorem 4. A generic fourth ray meets that quadric in two points and therefore touches exactly two members of the ruling — so the *minimum* gap between r_4 and the transversal family is essentially zero whether or not the tuple is co-chained. The correct 3D reading of $\lambda = \mu$ is universal: r_4 lies on a common chain with the others iff *every* member of the ruling meets it, i.e. iff r_4 lies in the quadric, in the same ruling as r_1, r_2, r_3. The honest scalar defect is the maximum gap over the family. Numerically (`check_chaincriterion.js`): for a co-chained tuple the max gap over a sampled ruling is 4×10^{-13}; perturbing t_4 by 20 units leaves a min gap of ~2 (the two accidental stabs) but a max gap of ~212. Any implementation of a co-chained test against 3D geometry should use the universal form.

A3. Theorem 5.3, corollary: the gauge trades shape for depth, at rate exactly g

The kernel of the action on ray space, written in the rig's affine chart, is the one-parameter family

```
g·(X, Y, Z) = ( X/W,  gY/W,  [g·Zv(Zh−Z) + Zh(Z−Zv)] / (D·W) ),
W = [g(Zh−Z) + (Z−Zv)] / D,
```

verified to fix both slits pointwise, to preserve every address (hence every ray as a set and every image pixel), and to satisfy the group law $\mathrm{gauge}(g_1)\circ\mathrm{gauge}(g_2) = \mathrm{gauge}(g_1g_2)$, all to $\sim10^{-12}$ (`check_depthgauge.js`).

The corollary: since W and the new depth depend on Z alone, frontal planes map to frontal planes, and within each frontal plane the gauge acts as the linear map diag(1/W, g/W). A frontal circle therefore remains a frontal ellipse with axis ratio exactly g. This is the precise sense in which the gauge pays for depth with shape, and it closes a loop with Theorem 7: the gauged scene produces the identical image, so nothing in the image certifies its depth — *unless* the target is known to be round, in which case the measured aspect ratio pins the gauge (an aspect ratio of a round target is only consistent with one g, namely 1) and the depth reading of Solver V2.1 is certified. The codec's roundness prior is not a convenience; it is exactly the datum that quotients out the kernel of Theorem 5.3. This deserves a sentence in the Corollaries of Theorem 7 and, in the patent narrative, is the clean statement of why the method measures something a single image provably does not carry.

A4. Theorem 8, additions to the commentary on the collapse

Write the walking rig as $Zh = Zv + h$. Three refinements of "the bi-cross-ratio's two components fuse," all in `check_pinholecollapse.js`.

(i) By A1, $c = Dz \cdot h$ identically, so the bow of every chain dies linearly in the separation; the straightening of demo 23 is first-order in h with a known constant.

(ii) For rays threading four fixed scene points P_i, the second component is exactly independent of h: $t_i = X_i \cdot h/(Z_i - Zv)$, and the common factor h cancels in the cross-ratio, so $\mu = (X_1/(Z_1 - Zv), \ldots)$ at every separation. Only λ migrates. The fusion is entirely one component walking over to meet the other; verified constant to nine decimals across $h \in [0.5, 100]$, with $|\lambda - \mu|$ dying at log–log slope 1.004.

(iii) The fusion requires the four rays to be coplanar with the limit point $O = (0, 0, Zv)$ — equivalently, the scene points to lie on a plane through O. On such a plane, $Y/(Z - Zv)$ is an affine function of $X/(Z - Zv)$, so the two limit cross-ratios agree. For generic points $|\lambda - \mu|$ plateaus at a nonzero value (≈2.11 in the test configuration), and correctly so: four non-coplanar concurrent rays retain a two-parameter invariant even in projective geometry. The collapsing camera fuses the invariants exactly on the tuples where the pinhole has only one to offer — the images of collinear points.

A5. The Seam §5, proposition: the norm is the miss

The statement "two rays that meet in space are null-separated in the image" admits a metric upgrade. For rays with addresses (s_1, t_1), (s_2, t_2), write $\delta = (\Delta s, \Delta t) \in \mathbb{R} \oplus \mathbb{R}$ for their difference and $N(\delta) = \Delta s \cdot \Delta t$ for the double-number norm. With feet $p_i = (0, s_i, Zv)$ and directions $d_i = (t_i, -s_i, D)$, one computes $d_1 \times d_2 = (D \cdot \Delta s, D \cdot \Delta t, s_1 t_2 - s_2 t_1)$ and $(p_1 - p_2) \cdot (d_1 \times d_2) = -D \cdot \Delta s \cdot \Delta t$, whence the distance between the two lines is

```
gap(r₁, r₂) = D·|Δs·Δt| / √( D²Δs² + D²Δt² + (s₁t₂ − s₂t₁)² ).
```

The numerator is $D \cdot |N(\delta)|$; the denominator is strictly positive for distinct rays. So the split-complex norm of the address difference *is* the physical miss, up to a positive factor — with the honest caveat that the factor depends on the configuration through the moment term $s_1 t_2 - s_2 t_1$ and not on δ alone, so the statement is exact as a vanishing locus ($\text{gap} = 0 \Leftrightarrow N(\delta) = 0$) and proportional pointwise rather than globally. The identity matches the generic line–line distance to 8.5×10^{-14} over 20,000 random pairs (`check_incidence.js`). Section 5's light cone is not a metaphor: the interval form of the picture's 1+1 geometry measures, in scene units, how badly two rays miss.

A6. The Seam §4, remark: the prohibition is Theorem 3's cross, as algebra

Force the three-point chain construction of Theorem 6b through two 1-parallel rays (s_1, t_1), (s_1, t_2) and a generic anchor. The resulting bidegree-(1, 1) relation satisfies $c \cdot s_1 + d = 0$ exactly: the pole of the would-be Möbius graph lands at the shared first coordinate, and the "chain" degenerates into the cross — one full pencil line through the shared foot together with the residual line through the anchor. The Minkowski parallelism of §4 (parallel points are the unjoinable ones) and the degeneration of case 3 in Theorem 3 are the same phenomenon seen from the two sides of the dictionary: in the image, a function cannot take two values at one argument; in space, the only "lines" whose images pass through both rays are the ones that pass through their common point, and those image to crosses, not chains. Verified exact in `check_incidence.js`.

Verification summary

Addendum	Script	Key figures
A1	check_bowing.js	asymptote identities to 9 decimals; t constant on touched line, exact
A2	check_chaincriterion.js	co-chained max gap 4e−13; off-chain min ≈ 2 vs max ≈ 212
A3	check_depthgauge.js	slits fixed 3e−14; image drift 1e−12; group law 1e−12; roundness(g=1.3) = 1.3000
A4	check_pinholecollapse.js	c = Dz·h exact; μ constant to 9 decimals; fusion slope 1.004; generic plateau 2.11
A5	check_incidence.js	gap identity 8.5e−14 over 20k pairs
A6	check_incidence.js	$c \cdot s_1 + d = 0$ exact

Provenance

Consistent with the main document's honesty clause: nothing above claims novelty over the classical structure. A1, A4, and A6 are elementary consequences of the bidegree-(1, 1) form of chains; A2 is the classical fact that a line meets a quadric twice, recorded because it is an easy trap; A3 is a direct reading of the kernel computation in Theorem 5's proof; A5 is a cross-product identity. Their value is the same as the document's: assembly — and, in A3's case, the exact articulation of why the codec's measurement escapes the group-theoretic obstruction.

Companion to demos 25–32 (see geometry_companion.html); all scripts run under node with no dependencies.